AP BIOLOGY TOPIC 7.3

Artificial Selection & Selective Breeding

UNIT 7: NATURAL SELECTION (13–20% EXAM WEIGHT)

EVO-1.F | **EVO-1.G** | **SELECTIVE BREEDING**

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 7.3 (ARTIFICIAL SELECTION & SELECTIVE BREEDING)

CED Core Vocabulary

Artificial Selection (Selective Breeding) — An evolutionary process in which humans intentionally choose which individual organisms reproduce based on desirable heritable phenotypes.

Trait-Focused — Only individuals expressing desired phenotypes are allowed to breed.

Rapid Change — Produces dramatic phenotypic divergence in far fewer generations than natural selection because humans eliminate waiting for slow environmental pressures.

Animal Breeding — Grey wolves (Canis lupus) domesticated into hundreds of dog breeds (Chihuahua to Great Dane) varying in size, coat, and herding/retrieving behavior; livestock...

Key Features of Artificial Selection — Intentional: Humans define the selection criteria (yield, size, temperament, aesthetics) rather than natural environmental survival.

Non-Random Mating — Controlled cross-breeding or inbreeding concentrates target alleles.

Agriculture — Wild teosinte grass bred into modern high-yield corn (Zea mays); crops bred for drought tolerance and pest resistance.

Horticulture — Ornamental plants bred for flower color, size, and symmetry.

§2. WHAT IS ARTIFICIAL SELECTION?

Human-Directed Evolution

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

What Is Artificial Selection? ➔ **Artificial Selection (Selective Breeding)** ➔ **Key Features of Artificial Selection**

- **Artificial Selection (Selective Breeding):** An evolutionary process in which **humans intentionally choose** which individual organisms reproduce based on desirable heritable phenotypes.
- **Key Features of Artificial Selection:**
  - **Intentional:** Humans define the selection criteria (yield, size, temperament, aesthetics) rather than natural environmental survival.
  - **Trait-Focused:** Only individuals expressing desired phenotypes are allowed to breed.
  - **Non-Random Mating:** Controlled cross-breeding or inbreeding concentrates target alleles.
  - **Rapid Change:** Produces dramatic phenotypic divergence in far fewer generations than natural selection because humans eliminate waiting for slow environmental pressures.

[EXAMPLE & BIOLOGICAL APPLICATION] **Classic Examples of Artificial Selection:**

- **Agriculture:** Wild teosinte grass bred into modern high-yield corn (*Zea mays*); crops bred for drought tolerance and pest resistance.
- **Animal Breeding:** Grey wolves (*Canis lupus*) domesticated into hundreds of dog breeds (Chihuahua to Great Dane) varying in size, coat, and herding/retrieving behavior; livestock bred for milk or meat yield.
- **Horticulture:** Ornamental plants bred for flower color, size, and symmetry.

§3. GENETIC CONSEQUENCES & TRADE-OFFS OF ARTIFICIAL SELECTION

Diversity & Inbreeding

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

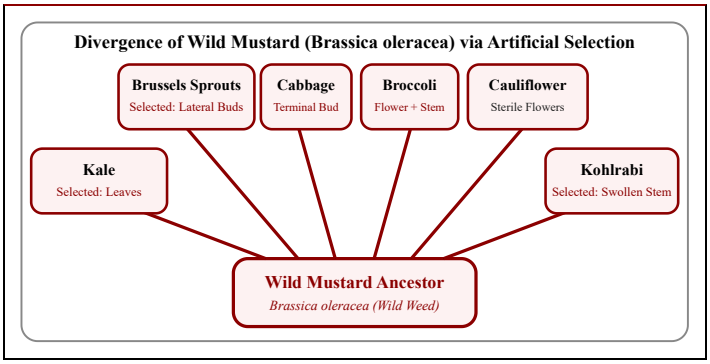
Genetic Consequences & Trade-Offs of Artificial Selection ➔ **Increases Frequency of Desired Alleles** ➔ **Reduces Overall Genetic Diversity** ➔ **Monoculture Vulnerability**

- **Increases Frequency of Desired Alleles:** Rapidly drives human-favored alleles toward fixation ($p \rightarrow 1.0$) while removing undesired alleles from the gene pool.
- **Reduces Overall Genetic Diversity:** Because only a small subset of individuals is bred each generation, genetic variation across the genome drops sharply (artificial bottleneck).
- **Inbreeding Depression & Deleterious Recessives:** Repeated mating among close relatives increases homozygosity ($q^2\uparrow$), exposing harmful recessive alleles that hitchhike alongside desired traits.
- **Monoculture Vulnerability:** Genetically uniform crops lack the genetic variation needed to resist novel pathogens or pests.

[EXAMPLE & BIOLOGICAL APPLICATION] **Trade-Offs in Purebred Dogs:** Selective breeding for physical traits has inadvertently concentrated harmful recessive disorders — e.g., hip dysplasia in German Shepherds, breathing problems in brachycephalic Bulldogs, and deafness in Dalmatians.

§5. ARTIFICIAL SELECTION VISUALIZED: BRASSICA OLERACEA

1 Ancestor → 6 Crops



By selecting for modifications of different plant organs in wild mustard, farmers created 6 distinct vegetables that are still the same biological species!

§6. ARTIFICIAL SELECTION VS. NATURAL SELECTION

Direct Comparison

FEATURE	ARTIFICIAL SELECTION	NATURAL SELECTION
Selective Agent	**Humans** (breeders, farmers, researchers)	**Environment** (predators, climate, competition, disease)
Traits Favored	Human-desired traits (yield, aesthetics, docility) — may *reduce* wild survival fitness	Adaptations that strictly increase **survival and reproductive fitness** in nature
Rate of Change	**Often very rapid** (visible changes in a few generations via strict breeding control)	**Usually gradual** (depends on strength of environmental selective pressure)
Mechanism	Both rely on **heritable phenotypic variation** and **differential reproduction** shifting allele frequencies	Both rely on **heritable phenotypic variation** and **differential reproduction** shifting allele frequencies

[AP EXAM FRQ SCORING STRATEGY] **Darwin's Analogy:** Darwin used artificial selection in fancy pigeons, dogs, and crops as experimental proof that selection can dramatically reshape species over time — if humans can do it in decades, nature can do far more over millions of years!

§4. CONVERGENT EVOLUTION: ENVIRONMENTAL SIMILARITY IN NATURE

EVO-1.G Connection

- Just as humans can select for similar traits across different breeds, **similar environmental selective pressures in nature** select for similar adaptations in distantly related lineages (**convergent evolution**).
- **Result:** Unrelated species independently evolve **analogous structures** that perform the same ecological function without sharing a recent common ancestor with that trait.

[EXAMPLE & BIOLOGICAL APPLICATION] **Australian Marsupials vs. Placental Mammals:** Sugar gliders (marsupial) and flying squirrels (placental) independently evolved gliding skin membranes because both occupy similar arboreal forest niches.

§7. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 7.3 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 7.3: Artificial Selection & Selective Breeding ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 1, 2 & 3 (Artificial Selection, Hardy-Weinberg Mathematical Modeling & BLAST Bioinformatics): Comparing allele frequency shifts across generations under directional selection vs. small-population genetic drift simulations, and constructing maximum-parsimony cladograms from BLAST percentage sequence identity matrices (Cytochrome c / β-globin). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Golden Rule: Hardy-Weinberg Equilibrium (HWE): Allele frequencies $p + q = 1$; Genotype frequencies p^2 (homozygous dominant) + $2pq$ (heterozygous carriers) + q^2 (homozygous recessive) = 1. Golden Calculation Sequence: ALWAYS begin with the recessive phenotype frequency (q^2), take $\sqrt{(q^2)}$ to find q, compute $p = 1 - q$, and then calculate $2pq$ (heterozygote carrier frequency). Never take the square root of the dominant phenotype frequency ($p^2 + 2pq$)! **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Evolutionary & Biomedical Disruption Case Study: Incomplete antibiotic regimens or agricultural antibiotic overuse exert strong directional selection on bacterial populations (e.g., MRSA or MDR-Tuberculosis): susceptible cells die while rare, pre-existing resistant mutants (arising via random mutation or R-plasmid conjugation) survive and reproduce. In conservation biology, severe population bottlenecks (e.g., Northern Elephant Seals, African Cheetahs) drastically erode genetic diversity via genetic drift, increasing homozygosity for deleterious recessive alleles and vulnerability to novel pathogens.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 7.3 to prove why the claim holds.

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